

TAP2901 Filter Change & Calibration Guide

Standard Operating Procedure for Filter Replacement & Optical Baseline (White Filter) Calibration

This document outlines the standard workflow for safely replacing the 47mm glass-fiber filter in the Brechtel TAP2901 and executing the automated optical baseline (White Filter) calibration via the DAQ User Interface.

Phase 1: Preparation & Hardware Intervention

Step 1.1: Enable Maintenance Mode

Before physically interacting with the instrument, the system must be halted to prevent erroneous data collection and safely stop the internal pump.

- Navigate to the instrument dashboard in your UI.
- Change the **Sampling State** setting to **maintenance**.
- *System Response:* The DAQ driver will automatically switch the instrument to Channel 0, stopping active sample flow and isolating the optical chamber.

Step 1.2: Physical Filter Replacement

With the flow safely stopped, proceed with the hardware intervention.

- Carefully remove the top hat and replace the internal 47mm filter. (*Refer to the Brechtel TAP2901 Manual for exact mechanical teardown, o-ring inspection, and reassembly steps*).
- **CRITICAL:** Install a HEPA / Zero-Air particulate filter onto the sample inlet of the TAP to ensure only particle-free air enters the chamber during calibration.
- Ensure the top hat is firmly secured and thumb screws are tightened in a star pattern.

Phase 2: Automated Calibration Sequence

Step 2.1: Initiate White Filter Routine

With the fresh filter and zero-air inlet installed, trigger the automated optical baseline script.

- In the instrument dashboard, change the **Calibration Routine** setting to `white_filter`.
- *System Response:* The DAQ driver will lock the instrument at Channel 0, clear internal buffers, and begin analyzing the raw optical intensities across all 8 spots and 3 wavelengths (Red, Green, Blue).

Step 2.2: Monitor Calibration Status

The DAQ driver autonomously manages the optical stability checks. Do not interfere with the instrument during this time.

- The **Calibration Status** field in the UI will display `in_progress`.
- The script monitors the optical drift over a 60-second rolling buffer. It will run until the Coefficient of Variation (CV) drops below 0.1% for all color channels.
- *Note: This process typically takes between 10 to 30 minutes, depending on the thermal stability of the instrument and the condition of the new filter.*

Phase 3: Completion & Return to Sampling

Step 3.1: Verify Calibration Results

The sequence concludes autonomously. Watch the UI for the following state transitions:

- **Success:** If the optics stabilize, **Calibration Status** will update to `success` and **Calibration Routine** will revert to `none`. The new baselines are permanently saved.
- **Timeout:** If the instrument fails to stabilize after 45 minutes, the driver forces an abort. **Calibration Status** will read `timeout_fallback`. The system will revert to the previous filter's baselines to prevent data corruption. *(If this occurs, check the hardware for leaks or thermal issues before attempting again).*

Step 3.2: Resume Ambient Sampling

Once the UI indicates the calibration routine has reverted to *none*, the instrument is ready to resume active measurement.

- **Remove the Zero-Air filter** from the sample inlet and reconnect the ambient sampling line.
- In the UI, change the **Sampling State** setting back to **sampling**.
- Change the **Active Spot** setting to **1** (or your preferred starting channel).
- *System Response:* The DAQ driver will engage the pump, actuate the requested spot, and resume real-time aerosol absorption logging.

Remember: Never switch the Sampling State back to "sampling" while the Calibration Routine is still reading "white_filter". Always wait for the automated sequence to finish and release its lock.